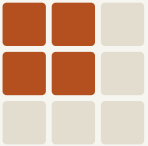
Slice Rank and the Cap Set Problem

Tao's polynomial method, explained from scratch

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Part 1 · A puzzle about lines

Warm-up: a card game hides our problem

In the card game **SET**, each of the 81 cards has 4 attributes, each taking one of 3 values.

Three cards form a "set" if, for **every** attribute, they are *all the same* or *all different*.

Question

What is the largest number of cards with no set among them?

Attribute	0	1	2
Number	one	two	three
Colour	red	green	purple
Shading	solid	striped	empty
Shape	oval	diamond	squiggle

A card is a vector, e.g. $(2, 0, 1, 1)$ = three, red, striped, diamond.
There are $3^4 = 81$ such vectors.

▶ Watch: a short video explaining SET

The field $\mathbb{F}_3$: arithmetic modulo 3

$\mathbb{F}_3 = \{0, 1, 2\}$: add and multiply as usual, then take the remainder mod 3.

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\times$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

Facts we will use

- $1 + 2 = 0$, so $-1 = 2$
- $t + t + t = 3t = 0$ for every t
- $t^2 = 1$ exactly when $t \neq 0$
- $t^3 = t$ for every t $(2^3 = 8 = 2)$

A **field** is a number system in which we can add, subtract, multiply, and divide by nonzero elements. $\mathbb{F}_3$ is one.

The space $\mathbb{F}_3^n$

A point of $\mathbb{F}_3^n$ is a vector $\mathbf{x} = (x_1, \ldots, x_n)$ with every $x_i \in \{0, 1, 2\}$. There are 3^n points.

We add vectors and multiply by scalars coordinate by coordinate, mod 3:

$$(1, 2) + (2, 2) = (0, 1), \qquad 2 \cdot (1, 2) = (2, 1).$$

The SET deck is exactly $\mathbb{F}_3^4$.

$(0,0)$	$(0,1)$	$(0,2)$
$(1,0)$	$(1,1)$	$(1,2)$
$(2,0)$	$(2,1)$	$(2,2)$

$\mathbb{F}_3^2$ as a 3×3 grid

(row = 1st coord., column = 2nd)

Lines in $\mathbb{F}_3^n$ have exactly three points

Definition

The line through $\mathbf{a}$ in direction $\mathbf{d} \neq \mathbf{0}$ is $\{\, \mathbf{a},\ \mathbf{a}+\mathbf{d},\ \mathbf{a}+2\mathbf{d} \,\}$ (it stops: $\mathbf{a}+3\mathbf{d} = \mathbf{a}$).



row, $\mathbf{d} = (0,1)$

$(0,0), (0,1), (0,2)$



column, $\mathbf{d} = (1,0)$

$(0,1), (1,1), (2,1)$



diagonal, $\mathbf{d} = (1,1)$

$(0,0), (1,1), (2,2)$



wrap-around, $\mathbf{d} = (1,1)$

$(0,1), (1,2), (2,0)$

$\mathbb{F}_3^2$ has 12 lines: 3 rows, 3 columns, 3 diagonals, 3 anti-diagonals (lines wrap around like on a torus).

Three ways to say “on a line”

Proposition

For distinct $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{F}_3^n$ the following are equivalent:

1. $\mathbf{x}, \mathbf{y}, \mathbf{z}$ lie on a common line;
2. in every coordinate i: $x_i = y_i = z_i$ *or* $\{x_i, y_i, z_i\} = \{0, 1, 2\}$;
3. $\mathbf{x} + \mathbf{y} + \mathbf{z} = \mathbf{0}$.

one coordinate (x_i, y_i, z_i)	sum	on a line?
all same (t, t, t)	$3t = 0$	allowed
all different $(0, 1, 2)$	$3 = 0$	allowed
two same (t, t, s), $s \neq t$	$2t + s \neq 0$	not allowed

(1)⇒(3):
$\mathbf{a} + (\mathbf{a} + \mathbf{d}) + (\mathbf{a} + 2\mathbf{d}) = 3\mathbf{a} + 3\mathbf{d} = \mathbf{0}$.

Condition (2) is exactly the rule of SET.

Two points determine the third:
$\mathbf{z} = -(\mathbf{x} + \mathbf{y})$.

Worked examples: add the three points

Example 1 in $\mathbb{F}_3^4$

coordinate	1	2	3	4
x	0	1	2	1
y	1	1	0	1
z	2	1	1	1
sum	**0**	**0**	**0**	**0**
pattern	diff.	same	diff.	same

$\mathbf{x}+\mathbf{y}+\mathbf{z}=\mathbf{0}$ $\Rightarrow$ **a line** ✓

Example 2 in $\mathbb{F}_3^2$

coordinate	1	2
x	0	0
y	1	0
z	1	1
sum	**2**	**1**
pattern	two same	two same

$\mathbf{x}+\mathbf{y}+\mathbf{z}=(2,1)\neq\mathbf{0}$ $\Rightarrow$ **not a line**

Two points decide the third: $-(\mathbf{x}+\mathbf{y})$. In Example 2 that is $(2,0)$, not $\mathbf{z}$.

Cap sets: no three points on a line

Definition

$S \subseteq \mathbb{F}_3^n$ is a **cap set** if there are no distinct $\mathbf{x}, \mathbf{y}, \mathbf{z} \in S$ with $\mathbf{x} + \mathbf{y} + \mathbf{z} = \mathbf{0}$.

$c(n)$ = the largest size of a cap set in $\mathbb{F}_3^n$.

$n = 1$**:** $c(1) = 2$



$\mathbb{F}_3$ is itself a line, so keep only 2 points, e.g. $\{0, 1\}$.

$n = 2$**:** $c(2) = 4$



$S = \{(0,0), (0,1), (1,0), (1,1)\}$. Check all 4 triples: none sums to $\mathbf{0}$. No 5-point cap exists (next slide).

DIY session: can 5 points avoid every line in $\mathbb{F}_3^2$?

Your turn

We found a cap set of size 4 in $\mathbb{F}_3^2$. Show that there is **no** cap set of size 5.

What is known for small n

n	1	2	3	4	5	6
$c(n)$	2	4	9	20	45	112
3^n	3	9	27	81	243	729
fraction $c(n)/3^n$	0.67	0.44	0.33	0.25	0.19	0.15

- $c(4) = 20$ answers the SET question (Pellegrino, 1970).
- $c(5) = 45$ (Edel, Ferret, Landjev, Storme, 2002); $c(6) = 112$ (Potechin, 2008).
- $c(7)$ is still **unknown**.

The fraction keeps shrinking. How fast?

A lower bound: $c(n) \geq 2^n$

Claim

$S = \{0,1\}^n$ (vectors with no coordinate equal to 2) is a cap set.

Proof. Take distinct $\mathbf{x}, \mathbf{y}, \mathbf{z} \in S$. They differ in some coordinate i. On a line, that coordinate would have to be *all different*, i.e. $\{x_i, y_i, z_i\} = \{0,1,2\}$, which needs a 2. Impossible. □

- So $2^n \leq c(n) \leq 3^n$.
- Cleverer constructions give $c(n) \geq 2.2^n$ for large n (best known around 2.22^n, partly found by computer search).

The big question

- **Meshulam (1995):** $c(n) \leq \dfrac{2 \cdot 3^n}{n}$ (Fourier analysis)
- **Bateman–Katz (2012):** $c(n) \leq \dfrac{3^n}{n^{1+\varepsilon}}$ (a long, hard paper)

Question (Frankl–Graham–Rödl, Alon, and others)

Is there a constant $K < 3$ with $c(n) \leq K^n$ for all n?

Terence Tao called it one of the most intriguing open problems in additive combinatorics.

2016: Croot–Lev–Pach (for $\mathbb{Z}_4^n$), then Ellenberg–Gijswijt: yes! A few pages, using polynomials. Tao then rephrased the proof via slice rank.

THEOREM (Ellenberg–Gijswijt, Annals of Math. 2017)

If $S \subseteq \mathbb{F}_3^n$ is a cap set, then

$$|S| \leq 2.756^n$$

So cap sets occupy a fraction at most $(2.756/3)^n \to 0$ *exponentially fast*.

Part 2 · Slices and slice rank

Functions of several variables ("tensors")

Fix a finite set X and a field $\mathbb{F}$ (think $\mathbb{F} = \mathbb{R}$ or $\mathbb{F}_3$).

Definition

A k-**tensor** on X is a function $f : X^k \to \mathbb{F}$.

We will mostly use $k = 2$ and $k = 3$: functions $f(x, y)$ or $f(x, y, z)$ with $x, y, z \in X$.

Notation: indicator function

$$\mathbf{1}_{x=a} = \begin{cases} 1 & \text{if } x = a, \\ 0 & \text{otherwise.} \end{cases}$$

Warm-up: the rank of a function of two variables

Definition

$f : X \times X \to \mathbb{F}$ has **rank one** if $f(x, y) = g(x)\, h(y)$ for some $g, h : X \to \mathbb{F}$.
$\operatorname{rank}(f) =$ the least r such that f is a sum of r rank-one functions.

Examples (over $\mathbb{R}$):

- $f(x, y) = xy^2 + 2y^2 = (x + 2) \cdot y^2$ has rank one.
- $f(x, y) = x + y = x \cdot 1 + 1 \cdot y$ has rank ≤ 2 (in fact exactly 2).

Easy upper bound: split by the value of x,

$$f(x, y) \;=\; \sum_{a \in X} \mathbf{1}_{x=a} \cdot f(a, y), \qquad \text{so } \operatorname{rank}(f) \leq |X|.$$

Diagonal functions have full rank

Fact

If $f(x,y) \neq 0$ only when $x = y$, then $\operatorname{rank}(f) \geq d$, where d is the number of a with $f(a,a) \neq 0$. In particular, if $f(x,y) \neq 0 \iff x = y$, then $\operatorname{rank}(f) = |X|$.

Proof. Let $D = \{a : f(a,a) \neq 0\}$ and suppose $f = \sum_{i=1}^{r} g_i(x)h_i(y)$ with $r < d$.

- Choose $v \neq 0$, supported on D, with $\sum_x v(x)g_i(x) = 0$ for all i (r equations, $d > r$ unknowns, so a nonzero solution exists).
- Then $\sum_x v(x)f(x,y) = \sum_i \left(\sum_x v(x)g_i(x)\right)h_i(y) = 0$ for every y.
- But the left side equals $v(y)f(y,y)$, which is nonzero when $v(y) \neq 0$. □

Key move: multiply by a well-chosen $v(x)$ and sum over x. Tao's lemma uses the same trick.

Slices

Definition

$f : X^3 \to \mathbb{F}$ is a **slice** if one variable can be *split off*:

$$f(x, y, z) = g(x)\, h(y, z) \quad \text{or} \quad g(y)\, h(x, z) \quad \text{or} \quad g(z)\, h(x, y).$$

In general, $f : X^k \to \mathbb{F}$ is a slice if $f = g(x_i) \cdot h(\text{the other } k-1 \text{ variables})$.

x	$y,\ z$	y	$x,\ z$	z	$x,\ y$

For $k = 2$ a slice is exactly a rank-one function $g(x)h(y)$.

Examples

A slice (over $\mathbb{R}$)

$x_1x_2 + x_2x_3^2$
$= x_2\,(x_1 + x_3^2)$

split off x_2 ✓

A slice (over $\mathbb{F}_2$)

$x_1x_2^3 + x_2$
$= x_1x_2 + x_2 = x_2(x_1 + 1)$

uses $t^3 = t$ in $\mathbb{F}_2$ ✓

Not a slice

$x_1x_2 + x_1x_3 + x_2x_3$

no single variable can be factored out (try to prove it!)

Also: any product $g(x)h(y)k(z)$ is a slice. The definition of a slice is *weaker* than "rank one": h may be any function of the remaining variables.

Slice rank

Definition (Tao, 2016)

The **slice rank** of $f : X^k \to \mathbb{F}$ is the least r such that $f = f_1 + \cdots + f_r$ with every f_i a slice.

Example. $f = x_1x_2 + x_1x_3 + x_2x_3$ is not a slice, so slice-rank$(f) \geq 2$. But

$$f = \underbrace{x_1\,(x_2 + x_3)}_{\text{slice in } x_1} + \underbrace{x_2\,(x_3)}_{\text{slice in } x_2},$$

so slice-rank$(f) = 2$.

- For $k = 2$, slice rank is the rank from the warm-up.
- Different slices may split off *different* variables.

An easy upper bound

Fact

For every $f : X^3 \to \mathbb{F}$, $\quad$ slice-rank$(f) \leq |X|$.

Proof. Split by the value of x:

$$f(x,y,z) \;=\; \sum_{a \in X} \underbrace{\mathbf{1}_{x=a}}_{g_a(x)} \cdot \underbrace{f(a,y,z)}_{h_a(y,z)}.$$

Each term is a slice in x. □

Example. $X = \{0,1,2\}$:

$$f(x,y,z) = \mathbf{1}_{x=0}\, f(0,y,z) + \mathbf{1}_{x=1}\, f(1,y,z) + \mathbf{1}_{x=2}\, f(2,y,z).$$

Tao's lemma: diagonal tensors have full slice rank

Lemma (Tao, 2016)

Suppose $f : X^3 \to \mathbb{F}$ is **diagonal**:

$$f(x, y, z) \neq 0 \iff x = y = z.$$

Then slice-rank$(f) = |X|$.

- "$\leq$" is the easy bound from the previous slide.
- "$\geq$" is the real content: you cannot do better than $|X|$ slices.
- Same idea as the two-variable fact: diagonal functions have full rank.
- The same statement holds for k-tensors, by induction on k.

A linear algebra tool

Support lemma

If $W \subseteq \mathbb{F}^X$ is a subspace of dimension d, then some $v \in W$ has at least d nonzero entries.

Proof. Put a basis of W in reduced row echelon form: basis vectors $w_1, \ldots, w_d$ with pivot positions $p_1, \ldots, p_d$, where w_j has a 1 at p_j and 0 at the other pivots. Then $v = w_1 + \cdots + w_d$ has entry 1 at every pivot. □

Example. $W =$ span of $(1, 0, 2, 1)$ and $(0, 1, 1, 0)$ in $\mathbb{F}_3^4$:
$v = (1, 1, 0, 1)$ has $3 \geq 2$ nonzero entries.

Proof of Tao's lemma (1/2): kill the x-slices

Suppose f is diagonal, $f(a,a,a) = c_a \neq 0$, and

$$f = \sum_{i=1}^{r_1} g_i(x)\, h_i(y,z) + \sum_{j=1}^{r_2} g'_j(y)\, h'_j(x,z) + \sum_{l=1}^{r_3} g''_l(z)\, h''_l(x,y)$$

with $r_1 + r_2 + r_3 < |X|$.

- $W = \{v : \sum_x v(x)\, g_i(x) = 0 \text{ for all } i \leq r_1\}$ is cut out by r_1 linear equations, so $\dim W \geq |X| - r_1$.
- By the support lemma, pick $v \in W$ with at least $|X| - r_1$ nonzero entries.
- Average f against v: $\quad F(y,z) = \sum_{x \in X} v(x)\, f(x,y,z)$.

Proof of Tao's lemma (2/2): compute $\operatorname{rank} F$ in two ways

Recall $F(y,z) = \sum_x v(x)\, f(x,y,z)$, a function of **two** variables.

Way 1: plug in the slices of f $\quad \Rightarrow \ \operatorname{rank} F \le r_2 + r_3$

x-slices:	$\left(\sum_x v(x)\, g_i(x)\right) h_i(y,z) = 0$	disappear
y-slices:	$g_j'(y) \cdot \left(\sum_x v(x)\, h_j'(x,z)\right)$	rank one each
z-slices:	$\left(\sum_x v(x)\, h_l''(x,y)\right) \cdot g_l''(z)$	rank one each

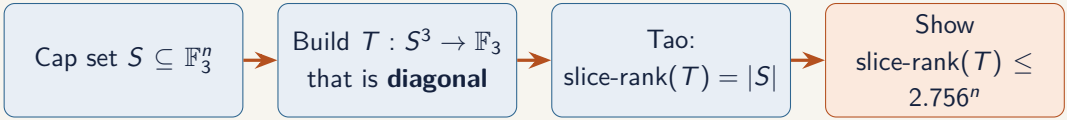
Way 2: use that f is diagonal $\quad \Rightarrow \ \operatorname{rank} F \ge |X| - r_1$

$F(y,z) = 0$ unless $y = z$, and $F(a,a) = v(a)\, c_a \neq 0$ whenever $v(a) \neq 0$.
That is at least $|X| - r_1$ values of a; apply the diagonal fact.

$$|X| - r_1 \ \le \ \operatorname{rank} F \ \le \ r_2 + r_3, \quad \text{so } |X| \le r_1 + r_2 + r_3. \text{ Contradiction!} \qquad \square$$

Part 3 · Proof of the cap set bound

The strategy



1. **Step 1.** Turn "no lines in S" into "T is diagonal".
2. **Step 2.** Write T as a polynomial and count slices.

Then $|S| = \text{slice-rank}(T) \leq$ (small number).

Step 1: a "zero detector" in $\mathbb{F}_3$

In $\mathbb{F}_3$, $t^2 = 1$ for $t \neq 0$. So

t	0	1	2
$1 - t^2$	1	0	0

Multiply over all coordinates:

The tensor

$$T(\mathbf{x}, \mathbf{y}, \mathbf{z}) \;=\; \prod_{i=1}^{n} \Big(1 - (x_i + y_i + z_i)^2\Big) \;=\; \begin{cases} 1 & \text{if } \mathbf{x} + \mathbf{y} + \mathbf{z} = \mathbf{0}, \\ 0 & \text{otherwise.} \end{cases}$$

(Equivalent to the form $\prod_i (s_i - 1)(s_i - 2)$ up to sign, since $(s-1)(s-2) = s^2 - 1$ in $\mathbb{F}_3$.)

Why T is diagonal on a cap set

Restrict T to $\mathbf{x}, \mathbf{y}, \mathbf{z} \in S$. Then $T(\mathbf{x}, \mathbf{y}, \mathbf{z}) \neq 0 \iff \mathbf{x} + \mathbf{y} + \mathbf{z} = \mathbf{0}$. Three cases:

Case	Is $\mathbf{x} + \mathbf{y} + \mathbf{z} = \mathbf{0}$?	T
$\mathbf{x} = \mathbf{y} = \mathbf{z}$	$3\mathbf{x} = \mathbf{0}$, yes	1
exactly two equal	$\mathbf{x} = \mathbf{y}$: then $\mathbf{z} = -2\mathbf{x} = \mathbf{x}$, impossible	0
all distinct	would be a line inside S, impossible	0

$T(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \mathbf{1}_{\mathbf{x}=\mathbf{y}=\mathbf{z}}$ on S^3. By Tao's lemma, slice-rank$(T) = |S|$.

Step 2: expand T into monomials

Case $n = 1$:

$$1 - (x + y + z)^2 = 1 - x^2 - y^2 - z^2 - 2xy - 2xz - 2yz.$$

Every monomial misses at least one variable, so it is a slice!

General n: each factor $1 - (x_i + y_i + z_i)^2$ has degree 2. Expanding the product gives monomials

$$\underbrace{x_1^{a_1} \cdots x_n^{a_n}}_{\mathbf{x^a}} \; \underbrace{y_1^{b_1} \cdots y_n^{b_n}}_{\mathbf{y^b}} \; \underbrace{z_1^{c_1} \cdots z_n^{c_n}}_{\mathbf{z^c}}, \qquad a_i + b_i + c_i \leq 2 \text{ for each } i.$$

So $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \{0, 1, 2\}^n$ and the total degree is $|\mathbf{a}| + |\mathbf{b}| + |\mathbf{c}| \leq 2n$, where $|\mathbf{a}| = a_1 + \cdots + a_n$.

The pigeonhole step

Since $|\mathbf{a}| + |\mathbf{b}| + |\mathbf{c}| \le 2n$, one of them is at most $2n/3$. Group each monomial by the first such variable:

$$T = \sum_{|\mathbf{a}| \le 2n/3} \mathbf{x}^{\mathbf{a}}\, F_{\mathbf{a}}(\mathbf{y}, \mathbf{z}) \;+\; \sum_{|\mathbf{b}| \le 2n/3} \mathbf{y}^{\mathbf{b}}\, G_{\mathbf{b}}(\mathbf{x}, \mathbf{z}) \;+\; \sum_{|\mathbf{c}| \le 2n/3} \mathbf{z}^{\mathbf{c}}\, H_{\mathbf{c}}(\mathbf{x}, \mathbf{y}).$$

Every term is a slice. Hence

$$|S| \;=\; \text{slice-rank}(T) \;\le\; 3\, M_n, \qquad M_n = \#\Big\{\mathbf{e} \in \{0,1,2\}^n : e_1 + \cdots + e_n \le \tfrac{2n}{3}\Big\}.$$

The degree, not the size of S, controls the number of slices. That is the whole trick.

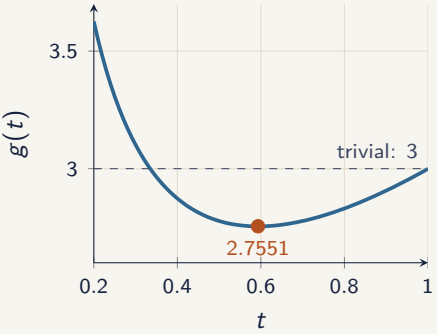
The exponential bound

Fix $0 < t \le 1$. If $e_1 + \cdots + e_n \le 2n/3$, then $t^{e_1+\cdots+e_n-2n/3} \ge 1$. So

$$M_n \le \sum_{\mathbf{e}\in\{0,1,2\}^n} t^{\sum e_i - 2n/3} = \left(\frac{1+t+t^2}{t^{2/3}}\right)^n.$$

Minimise $g(t) = (1+t+t^2)/t^{2/3}$: at $t \approx 0.5931$,

$$g(t) \approx 2.7551.$$



$|S| \le 3 \cdot 2.7551^n$.

(A Chernoff-type bound: few $\mathbf{e} \in \{0,1,2\}^n$ have sum far below the average n.)

Removing the factor 3: the tensor power trick

Observation

If $S \subseteq \mathbb{F}_3^n$ is a cap set, then $S^m = S \times \cdots \times S \subseteq \mathbb{F}_3^{nm}$ is a cap set.

Why: if $\mathbf{x} + \mathbf{y} + \mathbf{z} = \mathbf{0}$ in S^m, each block satisfies $\mathbf{x}^{(j)} + \mathbf{y}^{(j)} + \mathbf{z}^{(j)} = \mathbf{0}$ in S, which forces $\mathbf{x}^{(j)} = \mathbf{y}^{(j)} = \mathbf{z}^{(j)}$. So $\mathbf{x} = \mathbf{y} = \mathbf{z}$.

Apply the bound to S^m:

$$|S|^m \leq 3 \cdot 2.7551^{\,nm} \quad \Longrightarrow \quad |S| \leq 3^{1/m} \cdot 2.7551^{\,n} \xrightarrow{m \to \infty} 2.7551^{\,n}.$$

$|S| \leq 2.7551^n < 2.756^n$. $\square$

The whole proof on one slide

1. S has no lines $\Rightarrow$ $T(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \prod_i (1 - (x_i + y_i + z_i)^2)$ is **diagonal** on S^3.
2. Tao's lemma $\Rightarrow$ slice-rank$(T) = |S|$.
3. T is a polynomial of degree $\leq 2n$ $\Rightarrow$ each monomial has a variable group of degree $\leq 2n/3$.
4. Group monomials $\Rightarrow$ slice-rank$(T) \leq 3M_n \leq 3 \cdot 2.7551^n$.
5. Tensor power trick $\Rightarrow$ $|S| \leq 2.7551^n$.

Large set $\Rightarrow$ large slice rank. Low degree $\Rightarrow$ small slice rank.

Thank you!

Questions?